

VerbaClear: Project Documentation Summary

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Scope: B2B Live Event Accessibility & AI Vocabulary Assistant

1. Executive Summary & Core Vision

VerbaClear is an ambient, AI-powered real-time vocabulary assistant designed as a **B2B event technology solution**. The platform targets a deep psychological pain point: the public anxiety, cognitive overload, and loss of focus experienced by audience members when speakers use intense, advanced vocabulary or niche industry jargon.

Unlike traditional translation frameworks that output heavy, scrolling lines of raw text, VerbaClear works as an **educational intelligence filter** within the same language (English-to-Simple-English). It processes live speech invisibly, extracting only uncommon vocabulary and serving instantaneous, low-distraction synonyms to the entire room, preserving audience engagement and speaker flow.

2. System & Processing Architecture

The technical stack is built around a centralized, highly optimized local pipeline to eliminate high cloud computing costs and prioritize operational stability in live venues:

- **Audio Capture Layer:** Taps directly into the stage microphone, USB audio interface, or virtual loopback mixer.
- **Automatic Speech Recognition (ASR):** Utilizes locally embedded, pre-trained engines such as faster-whisper or Whisper.cpp. This removes real-time per-minute API costs, ensures complete data privacy, and enables fully offline operations.
- **Natural Language Processing (NLP) Filter:** Tokenizes incoming text streams in real-time and runs a string-matching comparison against a standard frequency database file like the New General Service List (NGSL). Any word spoken outside the common threshold (e.g., top 3,000–5,000 words) triggers processing.
- **Dual-Output Presentation Layer:** Pushes processed data instantly via WebSockets (e.g., Python FastAPI backend) to two unique frontend environments simultaneously.

3. Dual-Output UI Strategy

To optimize comprehension without causing visual distraction, VerbaClear segments data presentation into two specific tiers:

Feature Component	Primary Stage Presentation Screen	Audience QR Companion Portal
Target Intent	Subtle, split-second ambient glance.	Deep-dive contextual look and bookmarking.

Format Layout	Isolated lower-third overlay card featuring 1-3 punchy synonyms (e.g., <i>Labyrinthine</i> → <i>Highly Complex</i>). Fades out in 7 seconds.	Full mobile web environment displaying synonym + 1-sentence dictionary definition + phonetics.
Interaction	Zero interaction. Completely passive.	Active. Attendees tap to save words to a personal spaced-repetition deck (e.g., Anki).

4. Market Differentiation Matrix

Market research confirms a **complete blue ocean** for automated, event-wide vocabulary explanation software within a single language. Existing tools diverge fundamentally:

- **vs. Live Translators (e.g., Wordly.ai):** Translation engines bridge cross-lingual gaps (English to Swahili/French) for non-speakers, generating high cognitive fatigue. VerbaClear handles single-language gaps, translating dense English to accessible English.
- **vs. Traditional Captions (e.g., Zoom/Otter.ai):** Captions give equal screen weight to all words. VerbaClear extracts solely advanced vocabulary, leaving the screen clean and resting the audience's eyes on the speaker.

5. Commercialization & Business Scaling Architecture

VerbaClear scales smoothly as a B2B Software-as-a-Service (SaaS) model targeted at corporate events, universities, and large-scale venues with dedicated accessibility budgets:

1. **Tier 1 (Community / Non-Profit):** Time-capped processing access with a general vocabulary dictionary for community groups, youth events, or local public town halls.
2. **Tier 2 (Corporate & Academic SaaS):** Monthly or event-based licensing unlocking specialized **NLP Context Packs** (e.g., Legal, Medical, FinTech, Academic glossaries) and custom main-stage screen branding elements.
3. **Tier 3 (Enterprise AV Agency Integration):** High-premium licensing allowing massive audio-visual production houses to bundle VerbaClear natively into their physical corporate presentation stages and venue packages.

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